nature methods

Article

https://doi.org/10.1038/s41592-026-03034-9

3d-OT: a deep geometry-aware framework for heterogeneous slices alignment of spatial multi-omics

Received: 13 April 2025

Accepted: 17 February 2026

Published online: 27 March 2026

Check for updates

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The rapid advancement of spatial multi-omics technologies has unveiled opportunities for deciphering the intricate spatial heterogeneity; however, current computational approaches struggle to comprehensively integrate diverse molecular and spatial information. Here we propose 3d-OT, a deep geometry-aware framework that leverages spatial geometric and multi-omics information for feature extraction, spatial domains identification and heterogeneous slices alignment. 3d-OT utilizes modality fusion representation to align spatial slices, bridging the gap in spatial multi-omics alignment methods. Meanwhile, we handle nonrigid deformations in heterogeneous slice alignment through soft correspondence optimal transport, and the chamfer distance is introduced to quantify its performance. 3d-OT outperforms existing methods in capturing anatomical details of mouse brain cortex layers and tracking nonrigid deformations of heart and neural crest tissues at different resolutions. Finally, we construct the 3D spatiotemporal trajectory of mouse embryonic development. Overall, 3d-OT enables comprehensive understanding of existing spatial multi-omics data, offering a powerful computational tool to decipher the spatial complexity of biological tissues.

Advances in spatially resolved multi-omics have facilitated the exploration of intricate spatial heterogeneity. Compared to spatial transcriptomics technologies[1–6], spatial multi-omics technologies such as DBiT-seq[7], spatial CITE-seq[8], spatial ATAC–RNA-seq[9], spatial CUT&Tag–RNA-seq[9], SPOTS[10], SM-Omics[11] and spatial RNA-TCR-seq[12] provide distinct insights across various omics. By integrating complementary omics information, researchers can gain a more comprehensive and holistic understanding of cellular and tissue characteristics from diverse omics perspectives. In addition, spatial omics technologies typically involve multiple heterogeneous slices from different sources. Aligning these heterogeneous slices using spatial multi-omics information can reveal complex cellular relationships that cannot be detected within a single slice.

Several deep-learning methods[13–21] have been proposed to model spatial information for analysis of spatial omics data; however, we note that the application of spatial information is only to construct

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graph structures to capture dependency relationships between spatial locations.

Furthermore, nonrigid deformation and uneven resolution have always been challenges in alignment of heterogeneous slices[22], along with the absence of a universally accepted, objective alignment metric in various alignment methods[22–26]. As far as we know, there is currently no framework for the alignment of heterogeneous slices in spatial multi-omics. In addition to the problems above, many computational approaches for spatial omics data aim only at solving certain specific analytical tasks, while ignoring a scalable multi-task framework.

Here, we present 3d-OT, a deep geometry-aware framework capable of feature extraction, spatial domain identification and heterogeneous slice alignment. Unlike existing methods that rely on graph structures, 3d-OT adopts the point cloud segmentation model PointNet++[27], incorporating a mechanism that explicitly encodes positional information. Based on the input of fusion-modal representations, 3d-OT achieves the alignment of heterogeneous slices by employing a soft correspondence optimal transport (SCOT) strategy[28] with smooth loss and splat-based zero-divergence loss. Given the lack of a unified evaluation metric in alignment field, we introduce chamfer distance[29] to quantify the performance of 3d-OT.

We first demonstrated 3d-OT's performance in spatial clustering on simulated and experimentally acquired data of multiple samples including the human dorsolateral prefrontal cortex (DLPFC), human breast cancer, human lymph node and mouse brain coronal sections. By explicitly encoding positional information, 3d-OT captured more fine-grained anatomical details in cortex layers of mouse brain. We then evaluated 3d-OT alongside competing methods for aligning slices from the same platform (10x Visium[30], STARmap PLUS[31] and Stereo-seq[6]) as well as across different platforms (seq-FISH[32] and Stereo-seq). 3d-OT successfully tracked nonrigid deformation across varying resolution in complex structure, such as heart and neural crest tissues. We also reconstructed three-dimensional (3D) spatiotemporal differentiation trajectories for various tissues, including the liver and kidney, across different development stages. Our results showed that 3d-OT is a robust multi-task framework for thoroughly investigating the intricate cellular and molecular relationships within both existing and emerging spatial transcriptomics resources.

Results

Overview of the 3d-OT framework

3d-OT accomplishes fine-grained spatial domain segmentation and slice alignment for 3D reconstruction by integrating spatial multi-omics modalities data with spatially resolved information (Fig. 1a). Its geometry-aware deep-learning framework is composed of two core modules: (1) a PointNet++ Encoder and (2) a SCOT module. 3d-OT accepts input in the form of feature matrices and spatial coordinates derived from either segmented cells or spatial units, including beads, bins, pixels, spots and voxels. For simplicity, we will henceforth refer to cells and their captured coordinates as 'spots'. Leveraging its modular architecture, 3d-OT seamlessly supports both single-modality and multimodal tasks concurrently. Given the consistency of the overarching framework, our discussion here emphasizes the implementation of multimodal tasks.

For multi-omics data integration within a single spatial slice, 3d-OT used the PointNet++ Encoder to integrate geometric information and measurement features within a single omics and then between multi-omics integration. Within each omics, the spatial coordinates were utilized to construct a spatial graph and were integrated with the features before every layer of the PointNet++ (Fig. 1b,c). The different omics have distinct and complementary contributions to each spot, thereby we further implemented a multilayer perception (MLP) to generate the final multi-omics fusional representation. For the single-omics model, we used Transformer to obtain the final latent representation (Supplementary Fig. 1). These representations were then used to cluster spatial domains for uncovering biological associations. The fusional representation and single-omics reconstructed representation from different slices were then input to the SCOT module to achieve various alignment tasks. This workflow was adaptable to alignment of slices from cross-omics, multi-omics and single-omics datasets. Three-dimensional reconstruction of the tissue was generated from the resulting alignments of multiple pairs of slices.

In the SCOT module, feature representations derived from distinct slices are utilized to calculate a similarity matrix, which subsequently informs the computation of the transport cost. An entropy regularization method based on Sinkhorn[33] was implemented to obtain the optimal transport plan. We reconstructed the aligned spot positions based on the soft correspondence weights from optimal transport plan. Chamfer distance was used as the reconstruction loss to optimize the positions of the reconstructed spots, and a smoothness loss was applied to ensure a smooth transition of position changes between adjacent slices. Finally, considering the local cell-type differences in spatial slices and drawing inspiration from 3D particle tracking[28], we adopted a splat-based zero-divergence loss to obtain an alignment flow that follows biological processes. The construction of zero-divergence loss allowed for local complex spatial variations while maintaining overall alignment in the same direction (Supplementary Fig. 2a,b).

To assess the effectiveness of the proposed 3d-OT model, we first performed ablation studies on representative datasets to evaluate the contributions of the various components (Supplementary Figs. 3 and 4 and Supplementary Notes 1 and 2). We then assessed the importance of the PointNet++ Encoder and SCOT in spatial multi-omics alignment (Supplementary Fig. 5 and Supplementary Note 3). Finally, we analyzed the sensitivity of 3d-OT to the k parameter in the k-nearest neighbor (k-NN) graph construction (Supplementary Fig. 6), with the k values used in this study summarized in Supplementary Table 1.

Benchmarking 3d-OT and existing methods on spatial single-omics data domain identification

We first benchmarked 3d-OT's performance on spatial single-omics domain identification using experimentally acquired data with ground truth labels. The ground truth annotation allowed us to evaluate the performance of 3d-OT against competing methods using supervised metrics, namely, normalized mutual information (NMI) and adjusted Rand index (ARI). For comparison, we tested seven state-of-the-art methods: SEDR, stLearn, SpaGCN, GAAEST, STAGATE, GraphST and SpaNCMG.

In the DLPFC dataset, Maynard et al.[30] manually annotated the white matter and cortical layers (L1–L6) in 12 slices using gene markers and cellular structure (Supplementary Fig. 7). We visualized the annotations for the slice 151674 (Fig. 2a). The metrics confirmed that 3d-OT outperformed all methods across 12 slices, followed by SEDR, STAGETE, GAAEST, SpaNCMG and GraphST (Fig. 2b and Supplementary Figs. 7–9). 3d-OT was able to clearly recover the established understanding of cortical stratification in neuroscience. We visualized the clustering result of each approach, and found that SEDR, stLearn, SpaGCN and STAGATE only partially separated the hierarchical structures, whereas GAAEST, GraphST, SpaNCMG and 3d-OT exhibited better layering results. 3d-OT corresponded well with the manual annotation for all layers (Fig. 2c) and was the first method achieving ARI score greater than 0.7 for slices 151674 and 151673. Similar results were observed for slices 151671 (ARI = 0.849) and 151672 (ARI = 0.798), as shown in Supplementary Fig. 7.

We next examined the performance of 3d-OT with the same competing methods on the human breast cancer dataset. The 20 clusters were visualized based on the original analysis[34](Fig. 2d). 3d-OT achieved the best performance in the quantitative analysis of ARI and NMI (Fig. 2e), providing the best reconstruction of the annotated regions. 3d-OT was the only method that separated IDC_5 completely while maintaining the integrity of the Healthy_1_region (Fig. 2f).

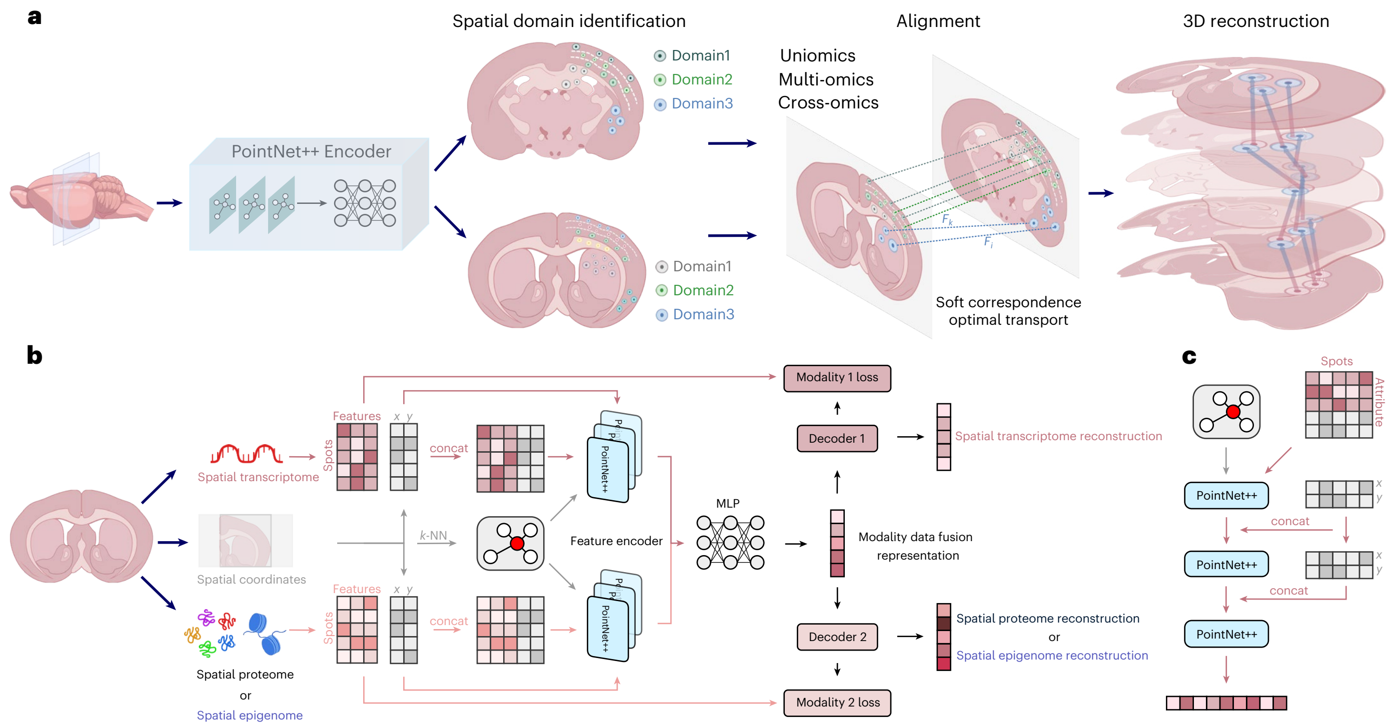

Fig. 1 | Architecture of 3d-OT multimodal framework. a, 3d-OT overview. 3d-OT is designed to leverage the spatial location information and expression data from multiple omics modalities across different spatial slices to achieve fine-grained spatial domain segmentation and accurate 3D reconstruction. **b**, PointNet++ Encoder structure. 3d-OT first uses the *k*-NN algorithm to construct a spatial neighbor graph using the spatial coordinates. Then for each omics, the preprocessed feature X and the spatial location information spatial $= (x, y)$ are concatenated into an attribute matrix and used as input to the PointNet++ Encoder. To capture the importance of different spot attributes, the spatial location information will be repeatedly concatenated into the potential representation. The modality-specific representations from different modalities are fused through an MLP to obtain the integrated representation of spots for fine-grained clusters. The integrated representation and the features of different modalities reconstructed through the decoder will serve as the input for the SCOT model to achieve alignment and 3D reconstruction tasks. **c**, PointNet++ model. Initially designed to capture local geometric information of point clouds, it improves the classification, segmentation and other geometric analysis tasks of 3D objects. Through its hierarchical structure, PointNet++ progressively extracts features from local to global scales. The attribute matrix after the two-dimensional spatial coordinates and feature matrix is spliced as input to each layer of PointNet++. Content from Generic Diagramming Platform (https://biogdp.com/) was used to draw the schematics (**a**,**b**)[61].

To further evaluate the single-omics performance of methods from previous comparisons, we used a mouse visual cortex dataset from the STARmap platform[35], which encompassed rich and detailed tissue information. We generated visualizations of the manual annotation on mouse visual cortex which contains the hippocampus, corpus callosum, layer 6 (L6), layer 5 (L5), layer 4 (L4), layer 2/3 (L2/3) and layer 1 (L1) (Fig. 2g). Visually, 3d-OT was able to clearly recover all seven hierarchical structures to closely match the ground truth (Fig. 2h). The metrics confirmed the visuals with 3d-OT scoring top in ARI and NMI, followed by STAGATE (Fig. 2i). 3d-OT clearly separated the five layers of the corpus callosum, L2/3, L4, L5 and L6 in the middle, whereas SpaNCMG and STAGATE were unable to distinguish between the L5 and L6 regions. GAAEST and SEDR could not separate the L2/3 and L4 regions (Fig. 2j). In addition, to further explore the applicability of 3d-OT, we integrated representations from the omics foundation model scGPT[36] for evaluation, but the performance did not exceed that of the original 3d-OT framework (Supplementary Fig. 10).

Benchmarking 3d-OT and existing methods on simulated and experimental spatial multi-omics data domain identification

To demonstrate 3d-OT's multi-omics domain identification performance, we benchmarked it with four competing methods: MISO, CellCharter, COSMOS and SpatialGlue using simulated data and experimentally acquired data. MISO was performed without hematoxylin and eosin (H&E)-stained images. The supervised metrics, namely, homogeneity, mutual information, v measure, AMI (adjusted mutual information), NMI and ARI, along with the unsupervised metric, Moran's I score, were used to quantitatively evaluate the performance of different methods.

We employed the same simulated data evaluation procedure as utilized in SpatialGlue, where factors model biological domains and modalities simulate the transcriptome and proteome, respectively. Modality 1 defines factors 1, 3 and 4 following the zero-inflated negative binomial distribution, whereas modality 2 determines factor 2 following the negative binomial distribution (Fig. 3a). 3d-OT achieved the best performance in the quantitative evaluation of six supervised metrics across five different parameter distributions of simulated data (Fig. 3b,c). Visually, 3d-OT clearly reconstructed the distribution of the four factors. Similarly, in the other four different parameter distributions of simulated data, 3d-OT exhibited the best performance (Supplementary Fig. 11).

We then benchmarked 3d-OT against four competing methods using a human lymph node dataset[20] (section A1) profiled with a 10x Visium RNA-protein co-assay. The original annotations are visually represented for reference (Fig. 3d). The annotation showed the major structures as the pericapsular adipose tissue and capsule, which formed the outer layers of the bulb, while the cortex and medulla (including sinuses, cords and vessels) constitute the core internal structures. 3d-OT exhibited the best performance on the evaluation metrics (Fig. 3e). All methods distinguished the cortex region, but only 3d-OT identified the capsule that forms the outer layers of the bulb. SpatialGlue and CellCharter were only able to partially identify the capsule region, whereas MISO and COSMOS failed entirely to detect it (Fig. 3f).

To further evaluate the performances of the best methods from previous comparisons, we used a spatial ATAC–RNA-seq[9] dataset

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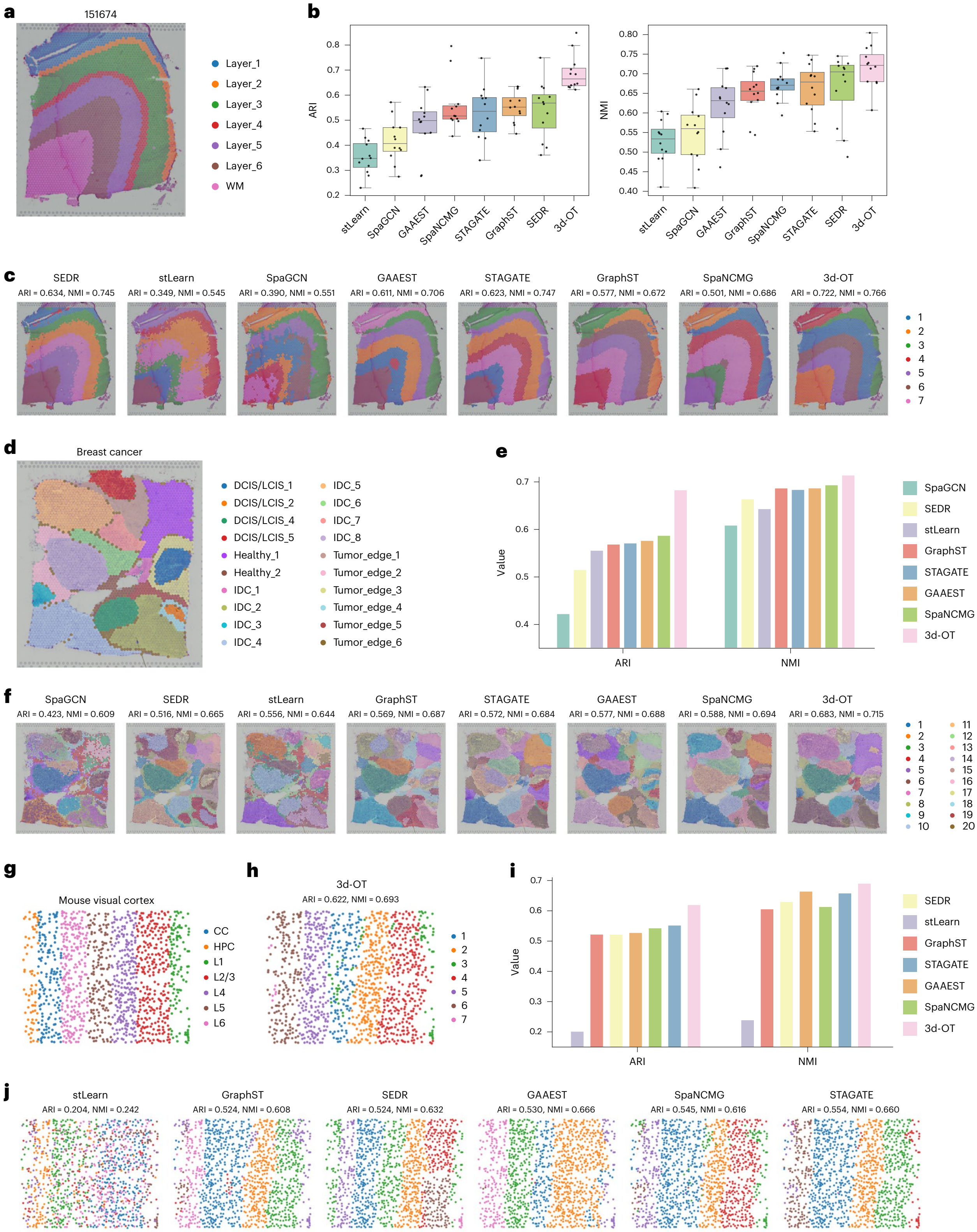

Fig. 2 | The performance of 3d-OT in single-omics spatial domain identification. **a**, Original annotation of 151674 slice in DLPFC dataset. WM, white matter. **b**, Boxplots of ARI values and NMI values for 3d-OT and 7 comparative methods on 12 slices. In the boxplot, the center line represents the median quartile, the box limits represent the upper and lower quartiles and the whiskers denote the 1.5× interquartile range. **c**, Visualization of the spatial clustering results for 3d-OT and 7 comparative methods on the 151674 slice. **d**, Original annotation of human breast cancer. DCIS, ductal carcinoma in situ; LCIS, lobular carcinoma in situ; IDC, invasive ductal carcinoma. **e**, Quantitative evaluation of eight methods with ARI and NMI. **f**, The clustering visualization of 3d-OT and 7 methods (cluster number = 20). **g**, Original annotation of mouse visual cortex. CC, corpus callosum; HPC, hippocampus. **h**, The clustering result of 3d-OT with ARI and NMI. **i**, Quantitative evaluation of 3d-OT and six methods. **j**, Visualization of spatial clustering results for six comparative methods.

comprising a slice of (P)22 mouse brain coronal section. We applied the same manual annotation, generated by COSMOS with reference to the Allen Brain Atlas, as a baseline for comparison (Fig. 3g). In terms of Moran's I score, 3d-OT ranked second, with COSMOS achieving the best result due to its lowest noise levels (Fig. 3h); however, 3d-OT demonstrated the highest performance on the ARI metric, indicating that the spatial domains identified by 3d-OT are more aligned with biological manual annotation (Fig. 3i). 3d-OT outperformed SpatialGlue and COSMOS in resolving distinct regions (L1–L3, L4, L5, L6a/b and caudoputamen), further confirming its superior ability in recovering the underlying biological truth of spatial data (Fig. 3j). To highlight the importance of multi-omics analysis, we showcased that 3d-OT, utilizing both modalities of mouse brain data, outperformed single-omics approaches that rely solely on transcriptomics data (Supplementary Fig. 12).

3d-OT dissects spatial epigenome–transcriptome mouse brain samples at higher resolution

We extended the analysis to another dataset of coronal sections incorporating a joint profiling of spatial CUT&Tag (H3K27ac histone modification) and spatial transcriptome (Fig. 4a). 3d-OT successfully segmented spatially distinct regions with similar transcriptional profiles (Fig. 4b), including cortical layer 6 subdomains (L6a/L6b) and lateral septal nucleus. These regions share similar gene expression signatures, especially cortex (ctx) layers 6a and 6b, which are often reported as one uniform layer[37]. COSMOS, though presenting the least noise, lacks accuracy when identifying the borders of each cluster, whereas SpatialGlue was unable to differentiate between cortical layers 6a and 6b (Fig. 4c,d and Supplementary Fig. 13a). The neighborhood enrichment results also demonstrated that 3d-OT outperformed SpatialGlue (Supplementary Fig. 13b). By explicitly integrating geometric information, 3d-OT successfully reflected a better anatomical distribution of the tissue.

The reason 3d-OT could distinguish these regions may be due to its higher sensitivity to marker genes compared to existing solutions. We listed the differentially expressed genes (DEGs) of each domain segmented by 3d-OT spatial CUT&Tag-RNA-seq result, and a large portion of the top-ranked genes were known markers of corresponding cell subtypes (Fig. 4e). Both *Gpr88* and the orthologs of *Pde1b* have been documented to distribute throughout the nucleus accumbens (ACB) and caudoputamen[38,39]. *Trpc4*, *Homer2* and *Dgkg* have all been reported as marker genes of LS neurons, and *Dgkg* has been highlighted to have a broader distribution in the septum, which is also consistent with our results[40]. *Ctgf*, *Cplx3* and *Nxph3* are all well-studied specific markers of cortex layer 6b[41–43], whereas *Rorb*, *Cux1* and *Cux2* spread widely across cortex layers[44,45]. *Rgs12* and *Pax6* have also been recorded to express in corresponding regions aligning with our results[46,47]. The predicted distribution of *Mbp* and *Sox2* was consistent with the original analysis, whereas *Nexn* (CP), *Cldn11* (ccg), *Fezf2* (ctx L6a), *Bcl11b* (CP) and *Tle4* (CP) were discovered to be potential markers by other domain segmentation methods[20,21]. Notably, three markers of L6b showed pronounced expression specificity confined to that particular layer; however, the weak expression signals of these genes likely explain why attention-based methods struggle to reliably identify this distinct domain. To quantify the ability of 3d-OT to identify fine-grained regions, we visualized the expression of three marker genes specific to the cortical layer 6b region (*Ctgf*, *Cplx3* and *Nxph3*). 3d-OT exhibited substantially enhanced specificity in detecting these genes in L6b (Supplementary Fig. 14). We then performed an ablation study to pinpoint the source of 3d-OT's enhanced resolution capability, revealing that it stems from both explicit geometric information and the PointNet++ architecture (Supplementary Fig. 15).

In addition to the reported genes, the visualization of these DEGs revealed several potential markers with highly specific spatial expression patterns: *Nefm* (ctx layers), *Pdyn* (ACB), *Nr4a2* (CLA/EPd) and *Garnl3* (ctx layer 6a). The precise delineation of cortical layer 6b has led us to hypothesize that *Garnl3*, a potential marker identified in layer 6a, may serve as a critical marker gene. *Garnl3* exhibited a specific spatial expression pattern in the L6a region of the brain, and the chromatin in this area is in a substantially open state, indicating that *Garnl3* may have high transcriptional activity. The L6a region exhibited abundant expressions and H3K27ac/H3K4me3 (active transcription-associated) but low H3K27me3 modifications, collectively supporting *Garnl3* as a potential L6a marker gene (Fig. 4f). The peak-to-gene links plots further demonstrated a strong enrichment of open chromatin and activating histone modifications in the promoter region of *Garnl3* (Extended Data Fig. 1a).

We then used the Homer package[48] to perform transcription factor binding site (Motif) enrichment analysis on the *Garnl3* promoter region. The known motifs of the transcription factors *Zbtb18* and *Zkscan1* were strongly enriched, indicating a potential regulatory relationship with *Garnl3* (Extended Data Fig. 1b). *Zbtb18* played a crucial role in the development of the mouse cerebral cortex by regulating neuronal migration and differentiation, ensuring the proper formation of cortical neural circuits[49,50]. Specifically, during the embryonic development of the mouse cerebral cortex, *Zbtb18* cooperated with other transcription factors, such as *Neurog2* and *Neurod1*, to ensure the correct migration of neurons from the ventricular zone to the cortical plate. This mechanism is crucial for establishing the layered structure and neural circuits of the mouse cerebral cortex.

To further evaluate the reliability of 3d-OT's reconstruction dealing with different modalities, we used another two spatial CUT&Tag-RNA-seq (H3K4me3 and H3K27me3 histone modification) datasets from the same source for domain segmentation (Supplementary Figs. 16 and 17). Based on the clustering results, COSMOS was unable to maintain its performance with different input modalities, whereas 3d-OT exhibited a high degree of consistency across four distinct datasets, providing robust evidence that it can effectively reconstruct the original anatomical regions with reliability across diverse modalities. SpatialGlue being the second-best method in retaining segmentation performance, we compared it to 3d-OT in neighbor enrichment analysis (Supplementary Figs. 16d and 17d), and the result demonstrated that our method has better spatial correlation. From the DEG visualization and neighbor enrichment analysis, it was evident that the 3d-OT's domain segmentation application can provide the most accurate anatomical truth about the input data across different modalities combination among all existing methods.

Benchmarking 3d-OT and existing aligning methods across distinct technologies and modalities

To evaluate the alignment performance of 3d-OT, we systematically benchmarked it against five leading methods (Spatial-Linked

Fig. 3 | The performance of 3d-OT in multi-omics spatial domain identification. **a**, Original annotation of simulated data 1. **b**, Boxplots of the six metrics with the scores from five simulated datasets. In the boxplot, the center line denotes the median, box limits denote the upper and lower quartiles, and whiskers denote the 1.5× the interquartile range. $n = 5$ simulated samples.
c, Visualization of the spatial clustering results for 3d-OT and four comparative methods. **d**, Original annotation of human lymph node sample A1. **e**, Quantitative evaluation of five methods with six supervised metrics. **f**, The clustering visualization of 3d-OT and four methods (cluster number = 10). **g**, Manual annotation of tissue from P22 mouse brain (spatial ATAC–RNA-seq). VL, lateral ventricle; ccg, genu of corpus callosum; aco, anterior commissure, olfactory limb. **h**, Boxplots of unsupervised metrics Moran's I score about the five methods. In the boxplot, the center line represents the median quartile, the box limits represent the upper and lower quartiles, and the whiskers denote the 1.5× interquartile range (cluster number = 18). **i**, Quantitative evaluation of five methods with supervised metrics ARI. **j**, The clustering visualization of 3d-OT and four methods (cluster number = 18).

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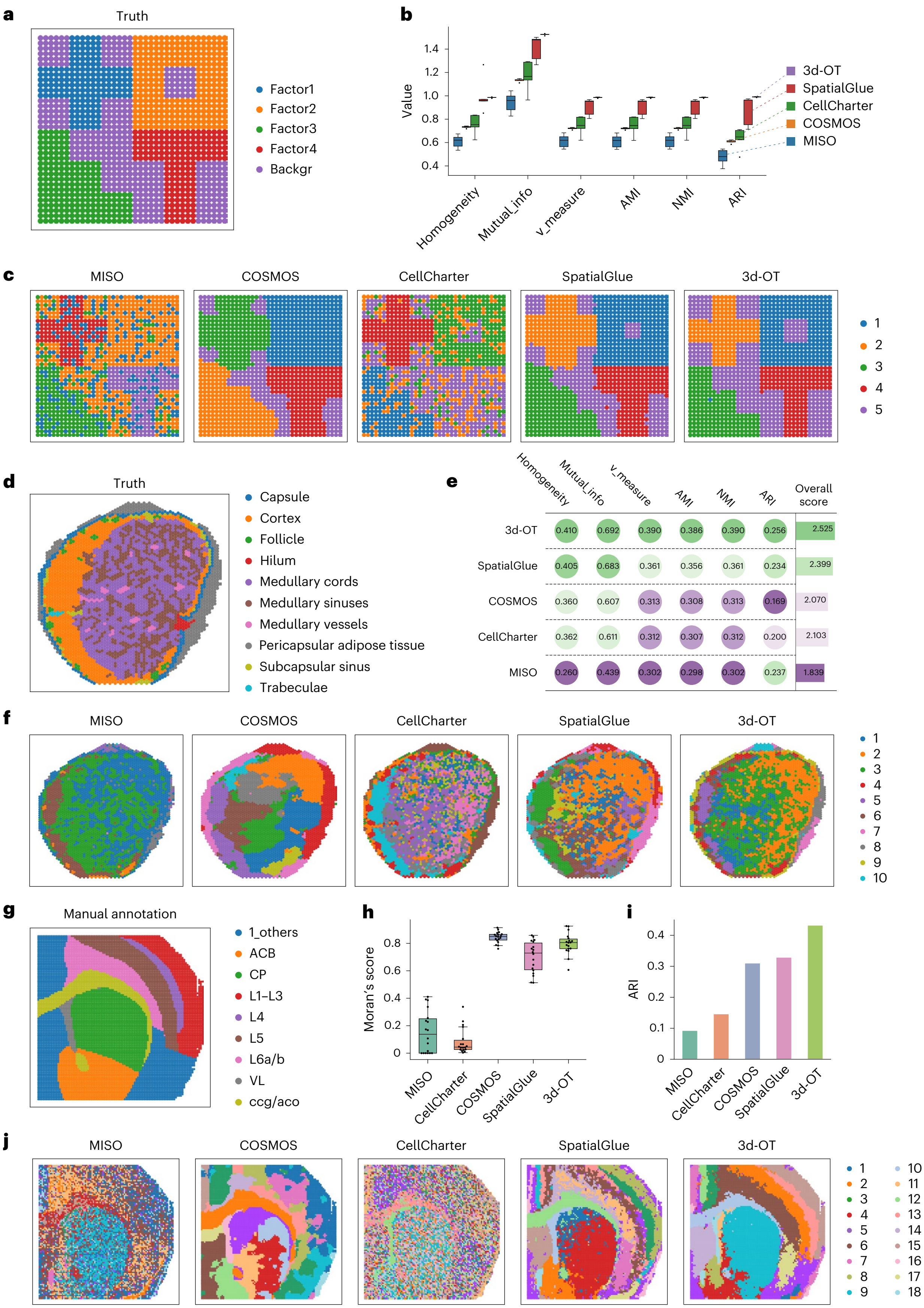

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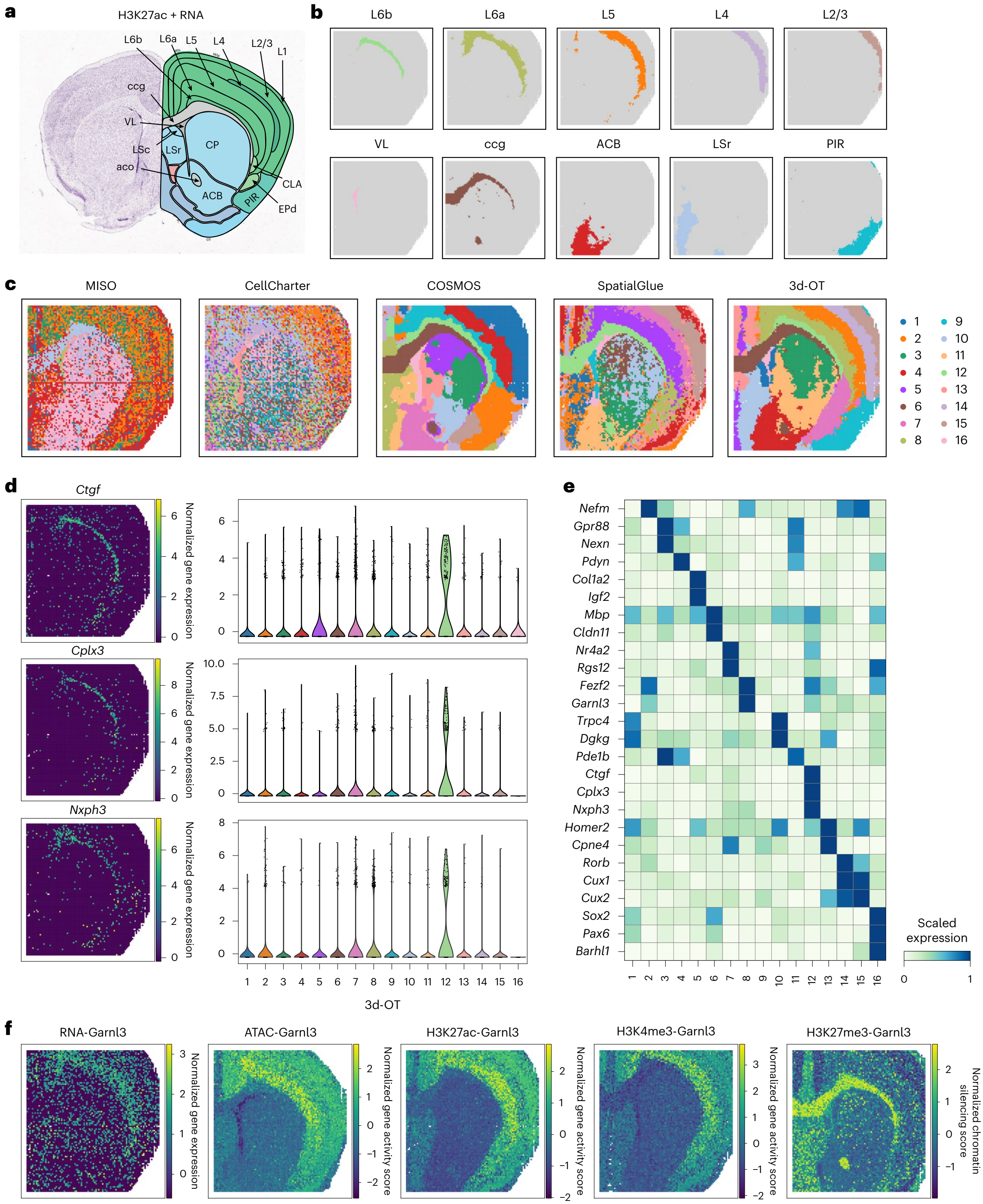

Fig. 4 | 3d-OT dissects spatial epigenome–transcriptome mouse brain samples at higher resolution. a, Annotation of mouse brain coronal sections from the Allen Brain Atlas. LSc, lateral septal nucleus, caudal part; LSr, lateral septal nucleus, rostral part; CLA, claustrum; EPd, endopiriform nucleus, dorsal part; PIR, piriform cortex; CP, caudoputamen. **b**, Separate spatial plots of ten annotated clusters identified by 3d-OT in the mouse brain (P)22 sample (spatial CUT&Tag-RNA-seq, H3K27ac). The annotated labels correspond to 3d-OT's results and the clustering colors do not necessarily match the same structures for the other methods. **c**, The clustering visualization of 3d-OT and four methods. **d**, Spatial mapping of marker genes expression for L6b. The violin plot colors correspond to 3d-OT's results and do not necessarily match the same structures for the other methods. Cluster 12 was annotated as L6b. **e**, Heatmap of markers expression for each cluster. **f**, Spatial mapping of Garnl3 expression for (P)22 mouse brain.

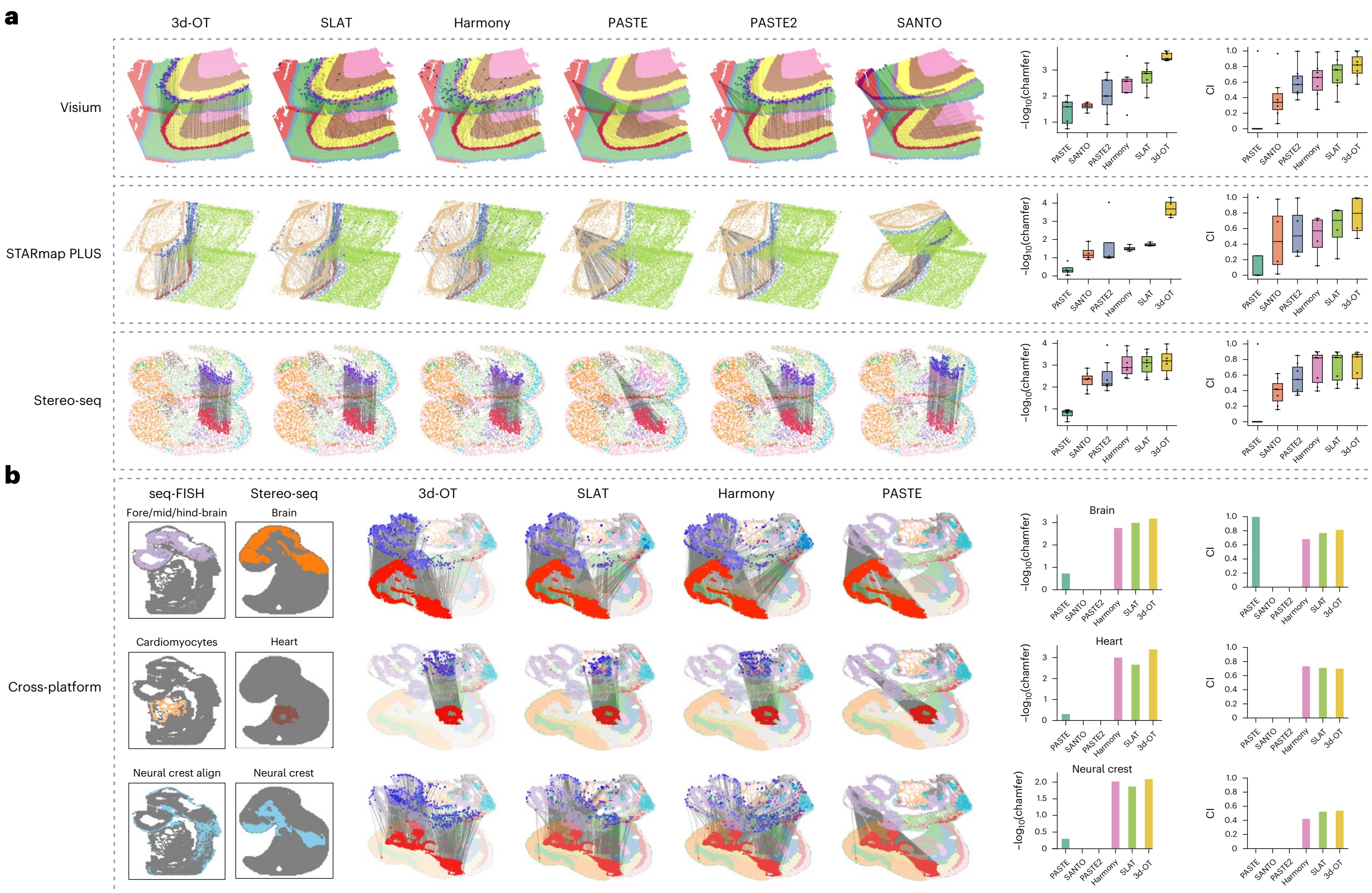

Fig. 5 | The performance of 3d-OT on the spatial alignment. a, Visualization of alignment results produced by 3d-OT and five comparative methods across three benchmark datasets. Boxplots show the distribution of two evaluation metrics: $\log_{10}$(chamfer distance) and CI for all six methods. In each boxplot, the center line indicates the median, the box limits correspond to the first and third quartiles and the whiskers represent 1.5× the interquartile range. **b**, Alignment visualization between E8.75 seq-FISH and E9.5 Stereo-seq mouse embryo datasets for 3d-OT, SLAT, Harmony and PASTE. Quantitative evaluation of six methods is presented using $-\log_{10}$(chamfer distance) and CI.

Alignment Tool (SLAT), Harmony, SANTO, PASTE and PASTE2). The evaluation for alignment tasks classified into the following three categories: within-platform alignment (three representative technologies, which reflect a wide array of methodology, resolution and throughput, specifically 10x Visium, STARmap PLUS and Stereo-seq), Cross-platform alignment (seq-FISH and Stereo-seq) and Multi-omics/Cross-omics alignment (spatial CUT&Tag-RNA-seq and ATAC–RNA-seq). We employ two metrics to evaluate the results: the chamfer distance and the Cell-Type Matching Index (CI)[22,26]. The chamfer distance, a well-established metric for assessing similarity between two-point clouds, is used to quantify the geometrical similarity between the aligned and original spot sets on the target slice.

We conducted the alignment benchmark using the 151673 and 151674 slices of DLPFC dataset and 3d-OT outperformed existing methods across all seven distinct regions (Fig. 5a and Supplementary Fig. 18). SLAT and Harmony ranked just behind 3d-OT. PASTE2, although it mitigated the alignment collapse issue[51] of PASTE, still failed to match the performance of the other three methods. 3d-OT also achieved the highest metric performance across the alveus, corpus callosum, cortex and hippocampus in an 8-month-old mouse brain dataset generated from STARmap PLUS technology (Fig. 5a and Supplementary Fig. 19). Next, we evaluated 3d-OT's capabilities with E15.5 mouse embryo Stereo-seq dataset and 3d-OT surpassed all others (Fig. 5a and Supplementary Fig. 20). We generated original annotations for two slices and selected seven different regions from them to assess the alignment performance, namely: brains, spinal cord, epidermis, liver, heart, muscle and connective tissue (Supplementary Fig. 20). The boxplot showed that 3d-OT achieved the highest median in the alignment results (Fig. 5a). Visually, both SLAT and Harmony show a sudden shift in position in the spinal cord region, whereas 3d-OT aligns this region smoothly. We subsequently evaluated the robustness of 3d-OT in scenarios lacking stereotypical spatial patterns. 3d-OT consistently preserved its performance when aligning an artificially fragmented DLPFC dataset and a human breast cancer dataset (Supplementary Fig. 21 and 22).

Spatial alignment across multiple technologies and modalities enables a more comprehensive cellular view[52]. To assess whether 3d-OT maintains consistent performance when analyzing the same biological samples across different technological platforms, we benchmarked the 3d-OT with other methods using both a seq-FISH E8.75 mouse embryo dataset and a Stereo-seq E9.5 mouse embryo dataset. Specifically, the seq-FISH dataset comprises over 10,000 cells, yet captures only ~350 genes, in contrast to the Stereo-seq dataset, which profiles approximately 5,000 cells and over 20,000 genes (Supplementary Fig. 23a). Alignment results revealed region-specific correspondences across platforms (Supplementary Fig. 23b), and we selected three corresponding regions (heart, cardiomyocytes; brain, fore/mid/hindbrain; neural crest, neural crest/cranial mesoderm/surface ectoderm) to evaluate cross-platform alignment performance of 3d-OT (Fig. 5b). SANTO and PASTE2 were omitted from the analysis due to their absence of cross-platform alignment functionality. We expanded the region corresponding to the neural crest in seq-FISH dataset to include the cranial mesoderm and surface

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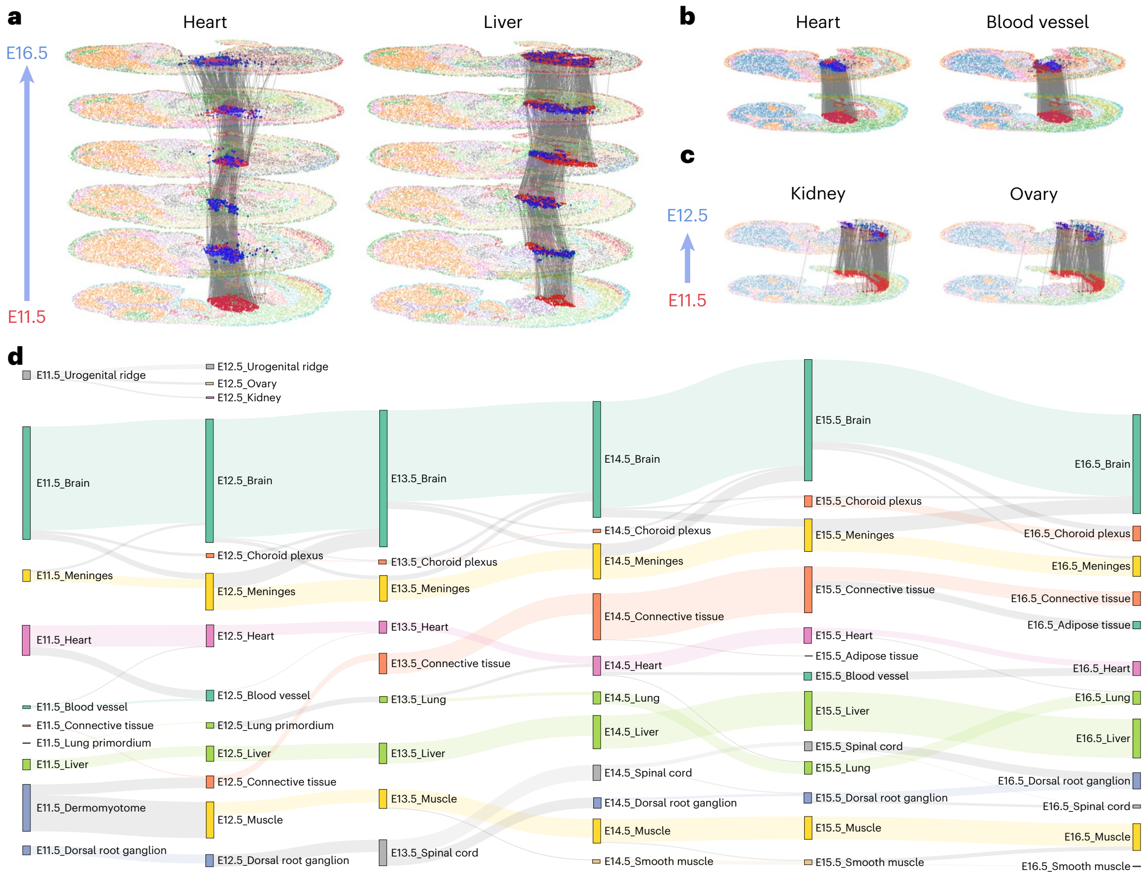

Fig. 6 | 3d-OT reproduces the development process of mouse embryo. **a**, Visualization of 3D reconstruction result of heart and liver from E11.5 days to E16.5 days. The blue spots represent the reconstruction spots from the SCOT model. **b**, Visualization of the alignment of E11.5 and E12.5 mouse embryo Stereo-seq datasets, colored by cell types. The red spot (bottom) represents the heart at E11.5 days. Red spot (top) represents the heart and blood vessel at E12.5 days respectively. The blue spot (top) represents the reconstruction spot from the SCOT model. **c**, Visualization of the alignment result of E11.5 urogenital ridge. The red spot (bottom) represents the urogenital ridge at E11.5 days. Red spot (top) represents the kidney and ovary at E12.5 days, respectively. The blue spot (top) represents the reconstruction spot from the SCOT model. **d**, Sankey plot of cell-type development from E11.5 to E16.5 as inferred from 3D reconstruction. Colored by the original annotation. Links containing less than 1% of cells are omitted for clear visualization.

ectoderm, which conforms with the approach adopted in the SLAT study. 3d-OT successfully tracked the nonrigid morphological deformations, while competing methods consistently produced results with physically inaccurate alignments (Fig. 5b). The quantitative evaluation results confirmed the visuals with 3d-OT scoring top.

Next, we test if 3d-OT can produce the same superior performance when applied to spatial slices of multi-omics data. We utilized two spatial slices from (P)22 mouse brain samples: spatial CUT&Tag-RNA-seq (RNA1 + H3K27ac) and spatial ATAC–RNA-seq (RNA2 + ATAC), as illustrated in Extended Data Figs. 2 and 3. The different omics information was assigned to five distinct sets (RNA1, RNA2; RNA1, ATAC; H3K27ac, RNA2; H3K27ac, ATAC; RNA1 + H3K27ac, ATAC + RNA2) to implement a comprehensive analysis (Supplementary Fig. 24). 3d-OT outperformed existing methods and was the only approach capable of addressing the multi-omics combination (RNA1 + H3K27ac; ATAC + RNA2). SLAT ranked second in the alignment task for five different combinations. As PASTE2 and SANTO were unable to achieve crossmodal alignment, we set their scores to ‘0’.

For a more comprehensive evaluation of 3d-OT’s multi-omics alignment capability, we conducted additional benchmarks on three sets of (P)22 mouse brain slices and on a multi-omics mouse spleen dataset[10] (Extended Data Fig. 3). Each set of brain slices comprised two slices with distinct epigenomic profiles (H3K27ac-H3K27me3, H3K27ac-H3K4me3 and H3K27ac-ATAC). Across these benchmarks, 3d-OT leveraging multi-omics representations outperformed all baseline methods, whereas 3d-OT using transcriptomics alone ranked immediately behind.

3d-OT constructed the spatiotemporal developmental trajectories of mouse embryo

Embryonic development is a dynamic and hierarchical process that encompasses the intricate formation of tissues and organs through a series of successive cellular events. We used 3d-OT to perform 3D reconstruction of the mouse embryonic developmental process from E11.5 to E16.5 and explored the spatiotemporal developmental trajectories of different tissues. We initially generated visualizations of the original annotations across six time points (Supplementary Fig. 25). Despite the positional shifts of the heart and liver throughout development posing challenges for nonrigid alignment across distinct time points, 3d-OT successfully captured their developmental trajectories (Fig. 6a). Visually, the red highlights denoted the positions of the heart and liver, while blue highlights indicated the locations of the reconstructed spots. It is noteworthy that the heart at E13.5 exhibits a ring-like structure, and 3d-OT successfully reconstructed the corresponding morphological distribution. The alignment flow distribution for the heart revealed a contraction from E11.5 to E14.5, followed by divergence from E14.5 to E16.5. Conversely, the alignment flow of the Liver exhibited continuous divergence. These observed patterns corresponded closely with the spatial distribution dynamics of the heart and liver at various developmental stages (Supplementary Fig. 25).

https://doi.org/10.1038/s41592-026-03034-9

To explore spatiotemporal dynamics during early mouse embryonic development, we analyzed the alignment results spanning E11.5 to E12.5. Our alignment results between the heart and blood vessel reflect their developmentally linked ontogeny (Fig. 6b), where blood vessels form adjacent to the early embryonic heart to establish circulatory support[53]. During cardiac development, vascular structures emerge within the heart, primarily through processes such as angiogenesis and vascular remodeling, ensuring an adequate blood supply to the developing organ[54]. We next followed up on kidney and ovary, two newly emerging organs at around E11.5–12.5. 3d-OT accurately identified them both as developing from the 'urogenital ridge' at E11.5 (refs. 55,56; Fig. 6c).

Finally, we generated the 3D reconstruction results of the major organs during the developmental process. The results indicated that most cells in the brain, heart and liver were well aligned with a high number of alignments, consistent with their spatiotemporal conservation at the six time points (Fig. 6d). In addition, several regions showed distinct differentiation phenomena, such as the dermomyotome development into muscle and connective tissue at E12.5, with muscle further development into smooth muscle at E14.5. During embryonic development (around E12.5), the somite-derived dermomyotome differentiates to give rise to both muscle and connective tissue[57,58]. Specifically, the myotome (dermomyotome-derived) generates muscle cells, while the dermatome gives rise to connective tissues including tendons and dermis. For some non-developmental alignment results, such as the alignment of the E12.5 dorsal root ganglion (DRG) with the spinal cord at E13.5, this might be due to the spinal cord information not being annotated in E12.5 (Supplementary Fig. 25). The DRG and the spinal cord had been in very close proximity to each other, in adjacent regions of the spine. The choroid plexus and the brain also exhibited a certain degree of alignment. During brain development, the choroid plexus occupies the lateral, third and fourth ventricles, serving distinct functional roles within these compartments. These results demonstrate that 3d-OT is highly effective in reconstructing embryonic developmental trajectories, facilitating the discovery of new differentiation pathways.

Discussion

3d-OT is a deep geometry-aware multi-task framework incorporating PointNet++ network and SCOT methodology. We designed 3d-OT to incorporate spatial geometric modality and multi-omics modalities by explicit utilization of positional information in the neural networks. With the presented examples, 3d-OT demonstrated a superior ability to utilize multimodal data for identifying domains in spatial slices. This capability can also be extended to align spatially resolved slices processed with different technologies or parameters. Our method proves the fact that existing methods have not extracted enough information from spatial location data. Furthermore, our quantitative benchmarking demonstrated that 3d-OT exhibited superior performance to seven state-of-the-art unimodal and four multimodal methods on 14 single-omics datasets and ten multi-omics datasets in spatial domain identification. The strong performance of 3d-OT underscores the necessity of a cohesive design, in which the incorporated explicit geometric information can be effectively captured by the PointNet++ architecture. In addition, the MLP-based modality fusion module facilitates the integration of complementary omics information, while the SCOT module further refines spatial alignment through targeted optimization of geometrical correspondences between slices. Collectively, these components enable 3d-OT to achieve robust and consistent performance across both spatial domain identification and slice alignment tasks. We applied the 3d-OT to a set of mouse brain epigenome–transcriptome data, revealing finer cortical layers and lateral septal nucleus regions. 3d-OT also discovered new potential marker genes in the L6a region. We also demonstrated the ability of 3d-OT to address heterogeneous alignment involving nonrigid deformation. This ability inspired us to construct a spatiotemporal trajectory mapping of mouse embryonic development.

To our knowledge, 3d-OT is the pioneering method that explicitly leverages geometric modalities and the first to accomplish both multimodal and crossmodal end-to-end alignment without relying on fused representations produced by external sources. Most existing computational spatial omics approaches aim only at addressing specific analytical tasks. Established methods like CellCharter, COSMOS, SpatialGlue and MISO are widely used for domain identification in spatial multi-omics. CellCharter effectively captures shared features across modalities by concatenating multiple omics into a single matrix for integrated embedding, but this approach may overlook modality-specific signals, potentially limiting its ability to fully leverage the unique contributions of each omics layer. COSMOS and SpatialGlue employ multi-encoder design to integrate multi-omics data, but their conventional graph convolutional network-based frameworks could be limited in capturing global geometrical features, leading to exaggerated delineations of certain regions in our benchmarking sections. MISO relies on H&E-stained histology images for optimal performance, limiting its applicability when such images are unavailable. In contrast, 3d-OT leverages a PointNet++ architecture with a data fusion strategy to robustly model global geometrical distributions while integrating diverse modalities, achieving strong performance across varied datasets.

Methods such as SLAT, SANTO, Harmony, PASTE and PASTE2 are well established for spatial slice alignment, but they were not specifically designed to align multi-omics slices. For methods that provide crossmodal alignment strategies or offer more flexible frameworks, we conducted crossmodal benchmarking, though several methods posed reproducibility challenges and could not be fully included in the comparison. In contrast, 3d-OT employs PointNet++ Encoders to explicitly capture both multi-omics and spatial information, followed by SCOT to achieve robust slice alignment even in the presence of pronounced differences between transcriptomic and epigenomic profiles. This capability is particularly relevant for advanced applications, such as reconstructing 3D tissue architectures, where crossmodal alignment is crucial for resolving complex biological structures. Currently, owing to the difficulty of collecting, spatial data that comprises joint profiling of multi-omics are relatively scarce. We believe that in the future, incorporating information from diverse omics to align heterogeneous slices will provide comprehensive new insights into the processes of biological development.

In validating the performance of 3d-OT, we also explored the use of omics foundation models to assess their potential for extending our framework, but no performance improvement was observed. While existing pretrained models capture rich transcriptional patterns, their lack of explicit spatial representation may constrain their ability to fully model spatial organization and domain boundaries. The key characteristic for a spatial omics foundation model is that it should natively support spatial coordinates as input. Moreover, using single-cell data and spatial omics data as joint pre-training may achieve better results, just like Nicheformer[59] and CellPLM[60]. For multimodal processing, as verified by 3d-OT, MLP fusion strategy can be used to jointly encode information of different modalities.

Moving forward, several aspects that warrant further exploration include: (1) we plan to extend 3d-OT to incorporate image data into multimodal integration. Most spatial omics technologies are accompanied by the generation of images, which contain important histological morphological features. (2) Stitching task. In addition to longitudinal alignment tasks, the task of stitching multiple slices from the same tissue cross-section is also crucial. In the future, while using multi-omics to analyze spatial domains, stitching multi-omics slices from the same cross-section will provide a more comprehensive tissue perspective. (3) Batch correction of potential multi-omics representations. Although the multi-omics latent representations of individual slices are effective for deciphering spatial domains, batch effects still exist between multi-omics latent representations from different slices, which impact the accuracy of multi-omics alignment tasks. (4) We first introduced the chamfer distance to evaluate macroscopic geometric structure preservation in spatial slice alignment. While the widely adopted CI reflects microscopic cell-label matching achieved by alignment methods, it lacks the ability to assess whether the aligned spots exhibit credible geometrical distributions. In the cross-platform alignment benchmark, the CI metric did not rank 3d-OT as the best-performing method, but it achieved a more accurate reconstruction of the geometrical correspondence between slices than the top-ranked approach; however, when the aligned slices exhibit highly similar geometrical structures, only use of chamfer distance may not effectively distinguish the relative performance of different methods. In the future, we plan to integrate the chamfer distance with additional metrics to establish a more comprehensive evaluation framework.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41592-026-03034-9.

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Methods

Data preprocessing

Single-modal data preprocessing. For spatial domain identification, the gene expression counts were log-transformed and normalized by library size via the SCANPY package[62]. The top 3,000 highly variable genes (HVGs) were selected and used as input to the encoder. For the alignment of spatial slices, we reference SLAT[22] and use SVD-based cross-dataset matrix decomposition strategy to correct inter-sample batch effects.

Multimodal data preprocessing. To preprocess the transcriptomic data, pixels expressing fewer than 200 genes and genes expressing fewer than 200 pixels were filtered out. Next, the gene expression counts were log-transformed and normalized by library size via the SCANPY package[62]. The top 3,000 HVGs were selected and used as input to principal-component analysis (PCA) for dimensionality reduction. For consistency with the chromatin peak data, the first 50 principal components were retained and used as input to the encoder. For the chromatin peak data, we used latent semantic indexing[63] to reduce the raw chromatin peak counts data to 50 dimensions. For the antibodies (ADT) data, we first filtered out genes expressed in fewer than ten spots. The filtered gene expression counts were then log-transformed and normalized by library size using the SCANPY package. Finally, the top 3,000 HVGs were selected and used as input for PCA. We applied centered log ratio normalization to the raw protein expression counts. PCA was then performed on the normalized data and the top 30 principal components were used as input to the encoder.

The 3d-OT framework

3d-OT is a new model based on multi-module integration, designed to leverage the spatial location information and expression data from multiple omics modalities across different spatial slices to achieve fine-grained spatial domain segmentation and accurate heterogeneous slice alignment. The model consists of two distinct modules: (1) PointNet++ Encoder; and (2) SCOT. Benefiting from the modular design, 3d-OT readily extends to spatial multi-omics data with more than two modalities.

PointNet++ Encoder

PointNet++ Encoder is based on the point cloud deep-learning model PointNet++. Initially designed to capture the local geometric information of point clouds, PointNet++ improved the classification, segmentation, and other geometric analysis tasks of 3D objects. Through its hierarchical structure, progressively extracted features from local to global scales. This approach effectively captured local gene expression and its spatial location correlations in spatial omics data. The integration of two-dimensional (2D) spatial information allowed the model to more accurately distinguish complex geometric structures and object boundaries.

Construction of neighbor graph. For each spatial omics slice $\mathcal{X}$, we constructed a spatial neighborhood graph $G = (V, E)$ using the k-NN algorithm, where V and E denote the sets of spots and edges, respectively. The adjacency matrix of G is denoted as A, with $A_{i,j} = 1$ if an edge exists between spots i and j, and 0 otherwise. Based on the expression feature matrix derived from each slice and the neighborhood indices provided by G, we constructed a local feature matrix $M^{(i)}$ for each spot x_i in $\mathcal{X}$. The j-th row, denoted as $M_j^{(i)} \in \mathbb{R}^{d_{\text{modality}}}$, represents the expression feature vector of the j-th nearest neighbor of spot x_i. By stacking all spot-specific neighborhood matrices, we obtained a global neighboring feature tensor $\mathbf{X} \in \mathbb{R}^{N\times k\times d_{\text{modality}}}$ where N denotes the total number of spots and d_{modality} represents the dimensionality of the omics features for the slice. This tensor served as the input for subsequent integrative analysis.

PointNet++ Encoder for individual modality. Different omics data exhibit distinct feature distributions. To encode these data into a latent space, we employed an encoder framework built upon the PointNet++ architecture. To incorporate spatial information, we concatenated the exact spatial coordinates $\text{spatial} = (x, y)$ of each neighboring spot to the corresponding local feature matrix $\mathbf{X}_i$, in which $\mathbf{X}_i$ represents the i-th matrix of the tensor $\mathbf{X}$. Formally, for each spot $i = 1, \ldots, N$, the input feature matrix for the l-th layer of the PointNet++ Encoder is defined as:

$$\mathbf{X}_i^l = \text{concat}\left(\mathbf{X}_i, (x, y)\right) \tag{1}$$

where l represents the layer index of the encoder and (x, y) are the vectors of spatial coordinates for the neighbors of spot i. This approach is conceptually similar to learning graph representations from the spatial graph G, which captures physical proximity between spots. By explicitly including spatial coordinates, the model is better able to perceive the overall distribution pattern of the spots, thereby integrating local expression features with tissue spatial structure.

A PointNet++ Encoder layer is composed of three successive convolutional layers, each followed by batch normalization, and a final max pooling operation that yields the latent representation. Specifically, after the preprocessing step for the l-th layer, the integrated attribute tensor $\mathbf{X}^l \in \mathbb{R}^{N\times k\times (d_{\text{modality}}+2)}$ is provided as input to the following transformation:

$$X^l = \text{MaxPooling}\left(\text{BN}\left(\text{LeakyReLU}\left(\text{Conv}\left(\mathbf{X}^l\right)\right)\right)\right) \tag{2}$$

where X^l denotes the latent representation produced by the l-th PointNet++ layer. In our 3d-OT framework, three PointNet++ Encoder layers are stacked, and the final representation of the m-th modality is denoted as $\text{modality}_m X^3$. The latent representations from different modalities are fed into an MLP to learn a unified representation:

$$Z_{\text{fused}} = \text{MLP}\left(\text{concat}\left(\text{modality}_1 X^3, \text{modality}_2 X^3\right)\right) \tag{3}$$

where Z_{fused} denotes the fused-modality representation output by the MLP and $\text{modality}_1 X^3$ and $\text{modality}_2 X^3$ represent the latent representations from different modalities.

Module training in PointNet++ Encoder. To enforce the learned latent representation to preserve the expression profiles from different modalities, we design an individual decoder for each modality to reverse the integrated representation Z_{fused} back into the normalized expression space. Specifically, by leveraging Z_{fused} from the between-modality MLP layer as input, the representations X_{output}^l for each modality $m \in \{1, 2\}$ generated by the decoder at the l-th layer are formulated as follows:

$$\text{modality}_m X_{\text{output}}^l = \sigma\left(X_{\text{output}}^{l-1} \text{modality}_m W^l + \text{modality}_m b^l\right) \tag{4}$$

where W^l and b^l correspond to the trainable modality-specific weights and biases at each layer, respectively, with σ denoting the Rectified Linear Unit (ReLU) activation function. The input to the first decoder layer is Z_{fused}, and the output of the final layer is X_{recon}^3. The PointNet++ Encoder module is trained to minimize the expression reconstruction loss:

$$\text{Loss} = \frac{1}{N}\sum_{N}^{i=1}\left(\text{modality}_1 X_i - \text{modality}_1 X_{\text{recon},i}^3\right)^2 + \frac{1}{N}\sum_{N}^{i=1}\left(\text{modality}_2 X_i - \text{modality}_2 X_{\text{recon},i}^3\right)^2 \tag{5}$$

where $\text{modality}_1 X$ and $\text{modality}_2 X$ represent the preprocessed original features of the modalities 1 and 2, respectively.

Alignment module

Inspired by previous research[64,65], we implemented an optimal transport-based framework to achieve soft correspondence alignment between source and target slices. Optimal transport provides a mathematical framework for aligning probability distributions, while preserving their geometric properties, addressing the challenge of redistributing mass from a source to a target distribution with minimal cost.

To make the spatial structure of the estimated distribution better match the target distribution, we employed a soft correspondence strategy. Rather than enforcing strict one-to-one correspondence between spatial locations, we leveraged chamfer distance within an unsupervised optimization process, prioritizing structural and spatial geometric properties as the optimization objective to achieve flexible and accurate alignment.

Solving optimal transport for soft correspondence. The optimal transport computation requires a feature matrix $\Phi \in \mathbb{R}^{n\times d}$, where n denotes the number of spatial spots and d is the feature dimension. The matrix Φ can be instantiated in two ways:

(1) As the fused representation:

$$\Phi = Z_{\text{fused}} \in \mathbb{R}^{n\times d} \tag{6}$$

which is obtained from the MLP data fusion module; or

(2) As the reconstructed single-modality representation:

$$\Phi = \text{modality}_m X^3_{\text{recon}} \in \mathbb{R}^{n\times d} \tag{7}$$

where $\text{modality}_m X^3_{\text{recon}}$ denotes the reconstructed representation of modality m after three successive PointNet++ Encoder layers.

Let $\Phi_{\mathcal{X}}$ and $\Phi_{\mathcal{Y}}$ represent the feature matrices extracted from two input spatial slices, $\mathcal{X}$ and $\mathcal{Y}$, respectively. We initiate by computing the spot-to-spot similarity matrix, given by:

$$S_{i,j} = \frac{\Phi_{x_i}\cdot \Phi_{y_j}^T}{\|\Phi_{x_i}\|_2 \, \|\Phi_{y_j}\|_2} \tag{8}$$

where Φ_{x_i} represents the feature of the i-th spot in $\Phi_{\mathcal{X}}$ and Φ_{y_j} represents the feature of the j-th spot in $\Phi_{\mathcal{Y}}$. For each source spot $x_i \in \mathcal{X}$, we assigned an equal mass of $\frac{1}{|\mathcal{X}|}$ to ensure normalization. We then considered its transport to the target spot $y_j \in \mathcal{Y}$ with the cost matrix defined as:

$$C_{i,j} = 1 - S_{i,j} \tag{9}$$

In this formulation, a higher cost corresponds to a lower similarity in the feature space. We then apply the entropy-regularized Sinkhorn algorithm[33] to identify the optimal transport plan T^*:

$$\begin{aligned} T^* = \underset{T\in\mathbb{R}^n_+}{\text{argmin}} \Bigg[& \sum_{i,j}\left(\exp\left(-\frac{C_{i,j}}{\epsilon}\right)\times \text{support}_{i,j}\right) T_{i,j} \\ & + \epsilon \times \frac{\gamma+\epsilon}{\gamma}\sum_{i,j} T_{i,j}(\log T_{i,j} - 1)\Bigg] \end{aligned} \tag{10}$$

where $\gamma = e^{\text{gamma}}$ and $\epsilon = e^{\text{epsilon}}$, with gamma and epsilon being learnable parameters optimized during training. The operator $\text{support}_{i,j} = \left(1 - \frac{\sigma}{\|S_{i,j}\|}\right)_+ \odot S_{i,j}$ acts as a thresholding function. Here, ϵ controls the strength of entropy regularization, while γ serves as an additional tuning factor, modulating the convergence behavior of the transport matrix by adjusting the step size of the Sinkhorn iterations.

The optimal transport plan T^* yields the soft correspondence weight of spot x_i to spot y_j:

$$W_{i,j} = \frac{e^{T^*_{i,j}}}{\sum_{k\in\mathcal{M}_{\mathcal{Y}}(x_i)} e^{T^*_{i,k}}} \tag{11}$$

where $\mathcal{M}_{\mathcal{Y}}(x_i)$ denotes the set of L candidate spots from $\mathcal{Y}$ corresponding to the top-L values of $T^*_{i,j}$, with L being a tunable hyperparameter. Based on these weights, the estimated position of spot y^*_j is obtained as:

$$y^*_j = \sum_{y_j\in\mathcal{Y}} W_{i,j} y_j \tag{12}$$

Finally, the alignment flow F_i from spot x_i to spot y_j is determined as:

$$F_i = y^*_j - x_i \tag{13}$$

Self-supervised losses. As manually linking spots between spatial slices is notably intricate, we adopt self-supervised loss function to train the alignment module.

Reconstruction loss. A core principle guiding self-supervised alignment flow learning is the fact that y^*_j and y_j should be similar. The chamfer distance is a standard metric used to measure the shape dissimilarity between point clouds in point cloud completion. We adopt it as our reconstruction loss:

$$L_{\text{recon}} = \frac{1}{|y'|}\sum_{y'_j\in y}\min_{y_j\in y}\| y'_j - y_j\|^2 \tag{14}$$

where y' represents the estimated spatial slice formed by y_j and y is the target spatial slice formed by y_j. By minimizing the chamfer distance, we ensure the consistency between the soft matching connection target and the estimated spatial slice location.

Smooth loss. The principle of smoothing loss is based on local consistency and continuity, ensuring that the alignment between adjacent slices remains smooth and no abrupt changes or discontinuities occur. Especially in spatial transcriptomics or image alignment problems, continuity of data and smooth transitions of spatial locations are crucial. The smooth loss is constructed as follows:

$$L_{\text{smooth}} = \sum_{x_i\in\mathcal{X}}\sum_{k\in\mathcal{N}_l(x_i)}\frac{\|F_i - F_k\|_1}{|\mathcal{X}|\,|\mathcal{N}_l(x_i)|} \tag{15}$$

where $\mathcal{X}$ represents the spatial slice formed by x_i, $\mathcal{N}_l(x_i)$ represents the index set of the l closest spots to x_i and F_i and F_k denote the estimated alignment flow at spot x_i and x_k, respectively. As shown in Supplementary Fig. 2. The alignment process is optimized by penalizing the alignment error between adjacent spots, ensuring that the spatial location changes smoothly throughout the alignment process.

Zero-divergence loss. Zero-divergence loss is similar to smooth loss in that it computes spatial gradients and requires the norm of the gradient to be small, essentially penalizing the case that neighboring alignment flow vectors are irrelevant; however, smooth regularization is too strict for spatial slice. While the divergence constraint only requires the total divergence to be zero, it does not necessitate that any two vectors be oriented in the same direction, thus allowing for locally intricate spatial location variations (Supplementary Fig. 2b). In practice, we set the neighborhood set size for calculating smooth loss to be much larger than that for calculating zero-divergence loss, because smoothness is a more coarse-scale regularization. Finally, inspired by 3D particle tracking research[28], we employ an efficient splatter-based method for calculating the zero-divergence loss.

Splat-based implementation. To compute divergence, we need the partial derivative of the field. The irregular arrangement of spots in 2D spatial slice complicates this. Thus, we improve upon previous[28], we splatting unstructured alignment flow estimates onto a uniform 2D grid, then applying zero-divergence regularization at these grid points, as shown in Supplementary Fig. 2c. In formal terms, the dense grid is denoted by $(sj, sk)^T$, with $j, k \in \mathbb{Z}$ indicating the 2D indices of the grid point. The parameters s correspond to the grids spacing. Then, given a grid, we employ the inverse squared distance as interpolation weights to approximate the alignment flow at that spot:

$$F(x) = \frac{1}{|N(x)|} \sum_{x_i \in N(x)} \frac{F_i}{\|x_i - x\|_2^2 + \epsilon} \tag{16}$$

where F_i is the estimated alignment flow value at spot x_i and $N(x)$ denotes the neighborhood among the spot set of spatial slices $\mathcal{X}$ for grid point x.

Divergence calculation. Once splatting has been employed, the divergence at that spot, specified by $x = (sj, sk)^T$, can be defined as:

$$(\nabla \cdot \mathbf{F})(x) = \sum_{k=1}^{2} \frac{\mathbf{f}(x + su_k) - \mathbf{f}(x - su_k)}{2s} \tag{17}$$

where u_k is a unit vector with 1 at the k-th entry. Finally, the zero-divergence regularization can be formulated as:

$$L_{\mathrm{div}} = \frac{1}{JK} \sum_{j=0}^{J-1} \sum_{k=0}^{K-1} \left\| (\nabla \cdot \mathbf{F})((sj, sk)^T) \right\|_1 \tag{18}$$

where J and K represent the number of grid points along the respective dimensions.

Therefore, the overall loss function used for alignment module training is defined according to:

$$L_{\mathrm{train}} = L_{\mathrm{recon}} + \lambda_{\mathrm{smooth}} L_{\mathrm{smooth}} + \lambda_{\mathrm{div}} L_{\mathrm{div}} \tag{19}$$

where $\lambda_{\mathrm{smooth}}$ and λ_{div} are weight factors that are utilized to balance the contribution of different loss.

Neighborhood enrichment. We calculated neighborhood enrichment with the Squidpy package[66] to assess the spatial relationships between clusters.

Chamfer distance. The chamfer distance is used to evaluate the similarity between two sets of points. It calculates the sum of the minimum squared Euclidean distances from each point in one set to the nearest point in the other set.

$$D_{\mathrm{chamfer}}(P, Q) = \frac{1}{|P|} \sum_{p \in P} \min_{q \in Q} \| p - q \|^2 + \frac{1}{|Q|} \sum_{q \in Q} \min_{p \in P} \| q - p \|^2 \tag{20}$$

where $P = \{p_1, p_2, \ldots, p_m\}$ and $Q = \{q_1, q_2, \ldots, q_n\}$ represent two sets of points. The distance is computed by summing the minimum squared distances between the points in the two sets and averaging over the number of points in each set. Specifically, we calculate the chamfer distance between the target aligned points and the actual aligned points. It is notable that 3d-OT still shows superior performance when using the reconstructed aligned position information instead of calculations based on the original position information (under the same number of alignments, calculations using the original coordinates result in a smaller chamfer distance).

Spatial clustering

Clustering was performed using the mclust R package via a Python–R interface. Specifically, mclust[67] was employed to identify clusters based on either the single-modality representation generated by the PointNet++ Encoder or the fused-modality representation integrated through the MLP module. The number of clusters was manually specified for benchmarking different methods. To capture known biological structures and cell types, we further iteratively tested different cluster numbers to achieve optimal performance.

3D reconstruction

We performed 3D reconstruction using the 3d-OT alignment pipeline. By pairwise alignment of closely related spatial transcriptomic slices, we establish a precise mapping of corresponding cells, reflecting developmental progression when samples differ primarily in ontogenetic stages. Through sequential alignment of a series of corresponding tissue slices representing distinct developmental stages within the same pipeline, we reconstruct comprehensive spatiotemporal developmental trajectories for the target organism.

Parameters of 3d-OT

3d-OT was trained for 1,150 epochs for single-omics domain identification. For the DLPFC, breast cancer and mouse visual cortex datasets, the top 3,000 spatially variable genes (SVGs) were selected, and the standardized expression matrix was used as input. The spatial graph was constructed with neighbors = 16. For multi-omics domain identification, training settings varied by dataset. For spatial ATAC–RNA-seq and spatial CUT&Tag–RNA-seq datasets, the model was trained for 300 epochs with neighbors = 16 and learning rate = 0.001. Simulated multi-omics datasets were trained for 600 epochs with neighbors = 16 and learning rate = 0.001. For the human lymph node A1 dataset, 500 epochs, neighbors = 6, and learning rate = 0.001 were used. These parameter choices were optimized to balance training stability and convergence speed.

In the alignment module of 3d-OT, a Euclidean distance-based threshold operator was applied to filter candidate end spots. For slices with relatively smooth spatial structure, the 50 nearest neighbors on the target slice were selected as candidates, from which the top five most similar spots were used to reconstruct the final end position. For slices with substantial resolution differences or weak spatial smoothness, 500–1,000 nearest neighbors were considered as candidates, and the top five most similar spots were retained for reconstruction. The alignment module was trained with the following parameters: learning rate = 0.0001, smooth loss weight = 1, divergence loss weight = 1, divergence neighbors = 8, smooth neighbors = 32 and maximum iteration steps in the Sinkhorn algorithm = 100.

Parameters of baseline methods

For all baseline methods, we adhered to the parameter settings recommended in their original publications whenever possible. For parameters not explicitly specified or for datasets requiring adaptation, we performed a limited grid search within a reasonable range to ensure competitive and fair performance against our proposed method.

GraphST was trained for 600 epochs with default parameters, using the full gene expression matrix as input.

STAGATE was trained for 1,000 epochs with standardized expression values of the top 3,000 SVGs, and the spatial graph was constructed with a radial cutoff of 150.

SpaGCN was trained for up to 200 epochs using the top 3,000 SVGs as input, with standardized expression values. The following parameters were specified: p = 0.5, start = 0.01, end = 1,000, tol = 0.01, max_run = 100, and r_seed = t_seed = n_seed = 200.

SpaNCMG was trained for 1,000 epochs using the top 3,000 SVGs with standardized expression values, with two spatial graphs constructed using rad_cutoff = 150 and k_cutoff = 6.

GAAEST was trained for 600 epochs using the full gene expression matrix, with hyperparameters set as $\alpha = 10$, $\beta = 1$, $\gamma = 1$ and $\lambda = 1$.

SEDR was trained for 200 epochs with the top 2,000 SVGs as input; the top 200 principal components after standardization were used, and the spatial graph was constructed with 12 neighbors.

SpatialGlue was trained for 1,600 epochs (neighbors = 3) in spatial ATAC–RNA-seq and spatial CUT&Tag–RNA-seq, for 200 epochs (neighbors = 6) in simulation data and for 200 epochs (neighbors = 6) in human lymph node.

COSMOS was trained with spatial_regularization_strength = 0.01, z_dim = 50, learning rate = 0.001, wnn_epoch = 500, total_epoch = 1,000, max_patience_bef = 10, max_patience_aft = 30, min_stop = 200, regularization_acceleration = True, edge_subset_sz = 1,000,000 and neighbors = 10.

For spatial slices alignment, SLAT constructed spatial graphs with k_cutoff = 10 and used feature = 'dpca' and join = 'inner' to account for batch effects. The resulting embeddings were subsequently used in spatial matching.

Harmony was run with join = 'inner', max_iter_harmony = 20, and PCA preprocessing, followed by spatial matching with the resulting embeddings.

PASTE was run with α = 0.1, while PASTE2 was run with α= 0.1 and s = 0.7.

SANTO was trained with learning rate = 0.001, 20 epochs, k = 10, α= 0.9, mode = 'align' and dimension=2.

Data and detailed methods
Details on the datasets, downstream analyses, competing methods and metrics used are available in Supplementary Tables 1–4.

Reporting summary
Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability
The 10x Visium human lymph node data were obtained from the Gene Expression Omnibus (GEO) under accession code GSE263617. The spatial epigenome–transcriptome mouse brain data were derived from AtlasXplore (https://web.atlasxomics.com/visualization/Fan/). The human breast cancer dataset is available at https://www.10xgenomics.com/datasets/human-breast-cancer-block-a-section-1-1-standard-1-1-0. The STARmap mouse visual cortex data were obtained from https://www.dropbox.com/sh/f7ebheru1lbz91s/AADm6D54GSEFXB1feRy6OSASa/visual_1020/20180505_BY3_1kgenes?dl=0&subfolder_nav_tracking=1. The STARmap PLUS 8 months mouse brain data were from https://singlecell.broadinstitute.org/single_cell/study/SCP1375. The Stereo-seq whole mouse embryo data were obtained from https://db.cngb.org/stomics/mosta/download/. The seq-FISH whole mouse embryo data were from https://crukci.shinyapps.io/SpatialMouseAtlas/. The multi-omics mouse spleen dataset was from GEO under accession code GSE263617. The data used as input to the methods tested in this study are freely available on Zenodo at https://zenodo.org/records/15089427 (ref 68). Source data are provided with this paper.

Code availability
The 3d-OT software package is available on GitHub at https://github.com/dbjzs/3d-OT. The Jupyter Notebook demonstrating how to reproduce our results and benchmarking experiments can be found at https://3d-ot.readthedocs.io/en/latest.

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Acknowledgements
This research was funded by the National Natural Scientific Foundation of China (no. 62571279 to Y.Z., no. 62501318 to H.L., no. 62303119 to Z.Y., no. 32470706 to Z.Y. and no. 62561045 to Z.F.), the Group Project of Developing Inner Mongolia through Talents (no. 2025TEL25 to Y.Z.), the Computational Biology Program of Science and Technology Commission of Shanghai Municipality (no. 25JS2850200 to Z.Y.), the Shanghai QiYuan Innovation Foundation (no. QY2025QNFDU009 to Z.Y.), the China Postdoctoral Science Foundation (no. 2024MD763987 to H.L.) and the Central Guidance Fund for Local Science and Technology Development (no. 2024ZY0168 to Y.Z.).

Author contributions
Y.Z. and Z.Y. conceptualized and supervised the project. B.D. designed and developed the method. B.D. and L.Y. prepared the figures and tables and contributed to the annotation of the P22 mouse brain dataset. B.D., L.Y., P.W., H.L., Z.F. and Y.Z. wrote and revised the paper. P.H., Y.S. and J.X. participated in the interpretation of results. All authors read, revised and approved the final paper.

Competing interests
The authors declare no competing interests.

Additional information
Extended data is available for this paper at https://doi.org/10.1038/s41592-026-03034-9.

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41592-026-03034-9.

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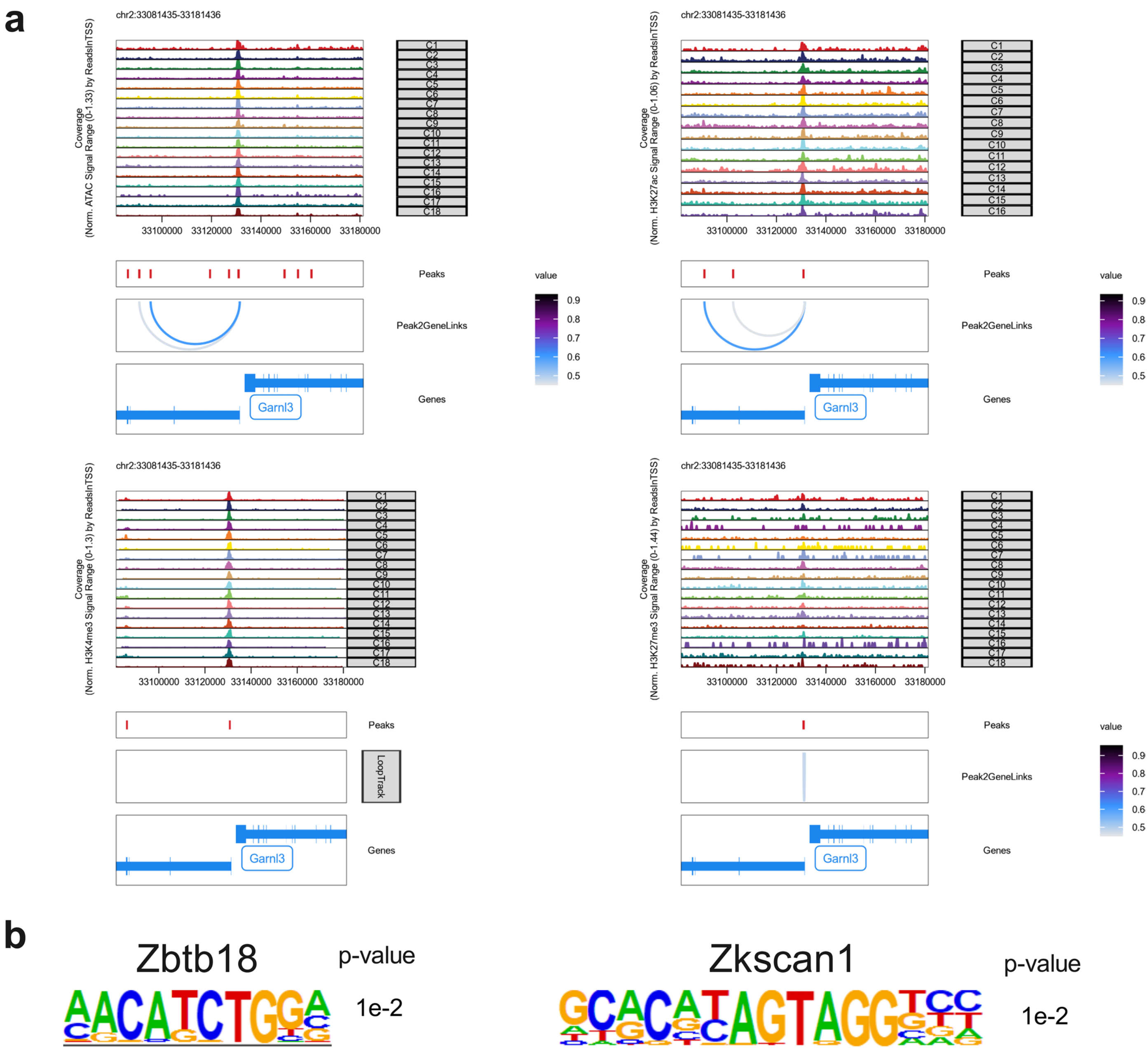
Peer review information *Nature Methods* thanks Lequan Yu, Weiqin Zhao, and other, anonymous reviewer(s) for their contribution to the peer review of this work. Peer reviewer reports are available. Primary Handling Editor: Madhura Mukhopadhyay, in collaboration with the *Nature Methods* team.

https://doi.org/10.1038/s41592-026-03034-9



Extended Data Fig. 1 | The epigenetic modification of *Garnl3*. **a**, Peak-to-gene links plots about *Garnl3* in ATAC, H3K27ac, H3K4me3, and H3K27me3. **b**, The enrichment results of known motifs for H3K27ac.

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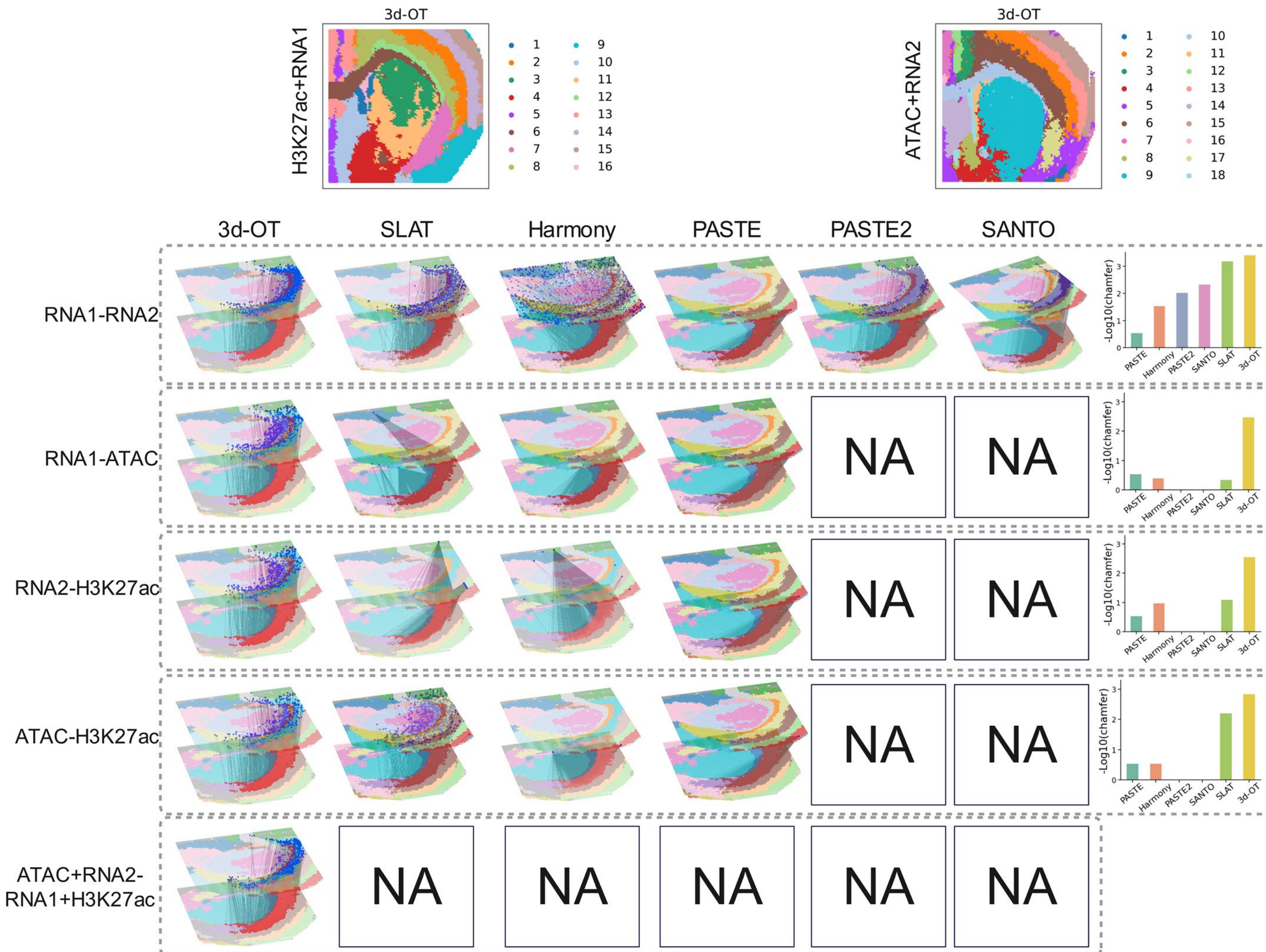
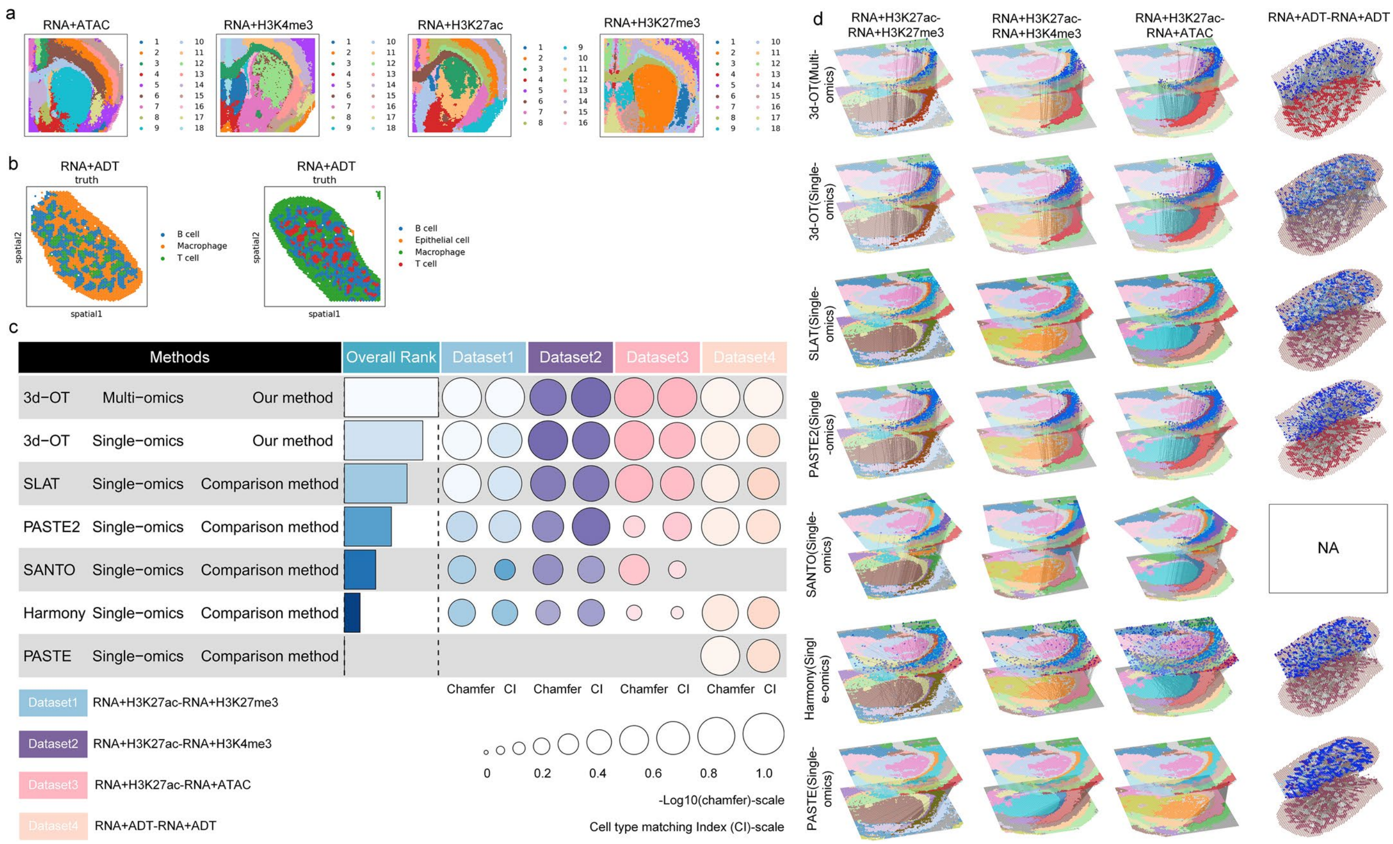


Extended Data Fig. 2 | Single-omics, cross-omics, and fusion-modal alignment of the (P)22 mouse brain slices. Visualization of the alignment between (P)22 mouse brain samples obtained via RNA-seq and CUT&Tag-seq (H3K27ac) and the spatial ATAC-RNA-seq sample. Since PASTE2 and SANTO cannot produce cross-modal alignment results, their outputs are marked as NA and assigned a value of 0 when calculating the chamfer distance. Notably, only 3d-OT supports the alignment of multimodal data fusion.

 https://doi.org/10.1038/s41592-026-03034-9



Extended Data Fig. 3 | Spatial slices alignment results using 3d-OT, SLAT, Harmony, PASTE, PASTE2, and SANTO on slices with multiple omics profile. **a**, Spatial clustering results of 3d-OT on each (P)22 mouse brain slices (RNA + ATAC; RNA+H3K4me3; RNA+H3K27ac; RNA+H3K27me3). **b**, Original annotation of mouse spleen dataset. **c**, Performance of each method on four datasets. The equation for the Overall Rank is: $rank_{overall} = 8/\sum(rank_{CI} + rank_{chamfer})$. For 3d-OT, we tested performance using both multi-omics and single-omics data, respectively, whereas other methods were evaluated using only single-omics data. **d**, Visualizations of the alignment results of each method on four datasets.

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Last updated by author(s): Nov 17, 2025

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Software and code

Policy information about availability of computer code

Data collection: No software was used for data collection.

Data analysis: The 3d-OT package is available at https://github.com/dbjzs/3d-OT. Usage of the code is introduced at https://3d-ot.readthedocs.io/en/latest/. Jupyter notebooks covering the analysis in this paper are available upon request. Specific package versions used in Python (3.10.13): numpy 1.26.3; scipy 1.12.0; scikit-learn 1.4.0; Scanpy 1.9.8; anndata 0.10.5.post1; torch 2.2.0; torch_geometric 2.4.0; libopenblas 0.3.25; scikit-misc==0.5.1; pandas 2.2.3; rpy2 3.5.11; squidpy 1.6.2. Specific package versions used in R (4.3.1): mclust 6.1.1; Hommer 5.1.0.

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The 10x Visium human lymph node data was obtained from the GEO (https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE263617).

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The Human breast cancer dataset is available at (https://www.10xgenomics.com/datasets/human-breast-cancer-block-a-section-1-1-standard-1-1-0).
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The data used as input to the methods tested in this study have been uploaded to Zenodo and are freely available at https://zenodo.org/records/15089427.

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Population characteristics	N.A.
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